difference between clustering and classification

classification is a supervised learning method where labeled data is used to train a model that assigns labels to new data. clustering is an unsupervised method where the goal is to discover natural groups without labels.

k-means clustering with k = 2

given points:

a(4,2), b(3,3), c(4,3), d(5,3), e(1,2), f(2,1), g(1,1), h(3,2)

initial centroids: (4,2) and (1,1)

pass 1 assignments:

cluster 0: a, b, c, d, h

cluster 1: e, f, g

centroids after pass 1:

cluster 0 centroid = (3.8, 2.6)

cluster 1 centroid = (1.3333, 1.3333)

within-cluster variation (wcv):

wcv = 5.3333

between-cluster variation (bcv):

bcv = 14.4174

ratio = bcv / wcv = 2.7031

initial ratio using initial centroids = 1.8021 $\rightarrow$ ratio increased after pass 1

pass 2:

assignments unchanged $\rightarrow$ algorithm converged

final clusters:

cluster 0: a(4,2), b(3,3), c(4,3), d(5,3), h(3,2)

cluster 1: e(1,2), f(2,1), g(1,1)

final centroids:

(3.8, 2.6) and (1.3333, 1.3333)